Topic 27 — Renewable & Non-Renewable Energy (अनवीकरणीय ऊर्जा)

Covers syllabus (click to expand)

Renewable | Non-renewable | Conventional | Non-conventional | Solar | Wind | Biomass | Biofuels | Biogas (बायोगैस) | Bioethanol (बायो-इथेनॉल) | Biodiesel (बायो-डीजल) | Small Hydro | Tidal | Ocean | Energy Security | India Scenario | National RE Programme | National Solar Mission | Energy Conservation | Energy Efficiency

> **Sources baked in:** NCERT Geo 11–12; MNRE; Energy Conservation Act 2001; BEE/PAT; National Policy on Biofuels 2018; National Solar Mission; UPPCS PYQs 2018–2025 **Weight:** ★★★ High — renewable vs non-renewable; biogas CH_4+CO_2; conventional/non-conventional; BEE/PAT; E20 **Last verified:** August 2026 **Overlap note:** Solar/wind/biomass/biogas/SHP/tidal detail also lives in the Renewable Energy (नवीकरणीय ऊर्जा) chapter — this file keeps those as one-home facts and expands **non-renewable, biofuels, ocean, security, conservation/efficiency**.

Current Affairs (this topic)

- Capacity and fuel-mix CA overlaps the Renewable Energy chapter — keep classification traps consistent.
- Nuclear is low-carbon but conventionally non-renewable in many Prelims frames.

27.1 Renewable Energy

Identity: Human-timescale replenishment | MNRE sources | Cleaner than fossils | Intermittency challenge

- **Renewable energy** comes from sources replenished continuously on a human timescale — sun, wind, flowing water, tides, biomass, geothermal heat.
- India types: solar, wind, biomass/biogas, small hydro, tidal, geothermal, waste-to-energy.
- Advantages: lower GHG than coal/oil; cuts oil/gas import pressure; decentralised rural supply.
- Limits: intermittency (night/no-wind); needs storage, hybrids, or backup; high upfront cost.
- **Energy crisis** commonly means the danger of exhaustion of **fossil fuels** such as coal and petroleum.
- UNGA declared **2014–2024** the **Decade of Sustainable Energy for All** (UN initiative supporting SDG 7).
- Ultimate energy in fossil fuels traces to the **Sun** via ancient photosynthesis.
- Among common options, **hydroelectricity** is often preferred for sustainable electricity generation versus coal, oil-gas, or nuclear stock depletion.
- India target: **500 GW non-fossil capacity by 2030** (COP26) — supports net zero **2070**.

- **Nuclear** is usually classified as **non-renewable** (finite uranium + waste).

> **Logic:** "Nuclear is renewable" = usually **FALSE** in UPPCS/UPSC (संघ लोक सेवा आयोग). Non-conventional ≈ renewable in many stems.

27.2 Non-renewable Energy

Identity: Coal, oil, gas, uranium | Finite stock | Oil import risk | Coal domestic

- **Non-renewable** sources form over geological time — use depletes the stock.
- **Coal:** India's primary electricity fuel; large domestic reserves in Jharkhand, Odisha, Chhattisgarh, West Bengal — eastern belt.
- **Petroleum:** India imports ~**85%** of crude — main energy-security vulnerability.
- **Natural gas:** cleaner than coal; used in power, fertiliser, CNG; partly imported (LNG).
- **Uranium/thorium:** nuclear fuel — low operational carbon, finite ore, radioactive waste — **not a fossil fuel**.
- Coal is **not** a biomass energy source — fossilised carbon, not living organic matter.
- Nuclear energy is **not** a biomass energy source.

PYQ — UPPCS Prelims 2021, Q124

A non-renewable source of energy is:

Options: A. Solar B. Petroleum C. Wind D. Biogas

Show answer

Ans: B — Petroleum is finite fossil fuel.

> **Logic:** India is **coal-rich domestically** but **oil-import dependent** — do not swap.

27.3 Conventional Sources of Energy

Identity: Coal, oil, gas, large hydro, nuclear | Traditional grid backbone

- In Indian usage, **conventional** means long-established grid sources: **coal, oil, natural gas, large hydropower (>25 MW), and nuclear**.
- Conventional ≠ only fossil — **large hydro and nuclear** are included.
- Advantages: baseload reliability and established infrastructure.
- Disadvantages: GHG (fossils), air pollution, mining/displacement, waste (nuclear).

Conventional	Non-conventional
Coal, oil, gas	Solar, wind, biomass

Conventional	Non-conventional
Large hydro	Small hydro ≤25 MW
Nuclear	Tidal, geothermal, ocean

27.4 Non-conventional Sources of Energy

Identity: Solar, wind, biomass, tidal, SHP, geothermal, ocean | ≈ renewable

- **Non-conventional** sources are newer/alternative sources — solar, wind, biomass, biogas, tidal, geothermal, ocean energy, small hydro.
- In MCQs, non-conventional sources are treated as **renewable**.

PYQ — UPPCS Prelims 2023, Q102

Non-conventional energy (गैर-पारंपरिक (वैकल्पिक) ऊर्जा स्रोत) sources are those that are—

Options: A. Non-renewable B. Renewable C. Produced from electricity D. Produced from heat

Show answer

Ans: B — Renewable.

> **Logic:** Non-conventional ≈ renewable; conventional includes fossils + large hydro + nuclear.

27.5 Solar Energy

Identity: PV + CSP (सीएसपी) | National Solar Mission 2010 | ISA Gurugram | 500 GW non-fossil 2030

- **PV** converts sunlight directly to electricity; **CSP** uses mirrors to heat fluid for steam turbines.
- **National Solar Mission (2010)** under NAPCC — solar capacity scaled toward ~100 GW by early 2020s.
- **ISA (2015):** India–France treaty IGO, HQ **Gurugram**; OSOWOG linked to ISA.
- **PM-KUSUM (2019):** solar pumps and farmer solarisation.
- Flagship park: **Bhadla (Rajasthan)**.

> **Logic:** ISA HQ = Gurugram. "38 members forever" outdated; OSOWOG under ISA is the tested true fact.

27.6 Wind Energy

Identity: Kinetic air → turbines | Power ∝ v^3 | Tamil Nadu leader | Onshore dominant

- Wind turbines convert kinetic energy of air to electricity; output rises with **cube of wind speed**.
- Leading states: **Tamil Nadu, Gujarat, Karnataka, Maharashtra**.
- Onshore mature; offshore potential on Gujarat/TN coasts — stronger winds, higher cost.
- Intermittent — needs grid balancing or hybrid solar–wind systems.

> **Logic:** Rajasthan leads **solar**, not wind.

27.7 Biomass Energy

Identity: Organic matter → heat/power | Bagasse UP sugar belt | Not coal/nuclear

- Biomass energy uses plant/animal organic matter — wood, dung, crop residue, bagasse.
- Direct combustion or conversion to biogas/biofuels.
- **Bagasse** cogeneration is major in western UP sugar mills.
- **Coal and nuclear are NOT biomass** sources.

PYQ — UPPCS Prelims 2022, Q69

NOT a biomass energy source:

Options: A. Coal B. Nuclear C. Gobar D. Wood

Show answer

Ans: A (also nuclear is not biomass) — coal is fossil; gobar/wood are biomass.

27.8 Biofuels

Identity: Liquid/gaseous fuels from biomass | Policy 2018 | E20 target | Food vs fuel debate

- **Biofuels** are fuels produced from biomass — bioethanol, biodiesel, biogas/CBG, advanced (2G) fuels.
- **National Policy on Biofuels, 2018** sets blending roadmap and feedstock categories.
- **1st generation:** sugarcane molasses ethanol; edible-oil biodiesel — food vs fuel concern.
- **2G:** agri-waste, residues — reduces food competition.
- Benefits: lower net GHG than fossils; domestic feedstock; energy security. Trap: biofuels are often **not cost-effective** versus fossil fuels.
- **Maize** is a classic **ethanol** crop; **Jatropha / Pongamia / sunflower** are classic **biodiesel** crops — do not swap.
- Biomass **gasification** (coconut/groundnut shells, rice husk) produces combustible **producer gas** (CO + H_2 + traces of CH_4) usable for heat and **internal combustion engines** — not "CO_2 only" and not "heat only."
- UP western sugarcane belt is a major **bioethanol** feedstock zone.

Fuel	Feedstock	Blend use
Bioethanol	Molasses, maize, 2G waste	Petrol E10/E20
Biodiesel	Jatropha, Pongamia, UCO	Diesel B5/B10
Biogas/CBG	Dung, organic waste	Cooking / transport

> **Logic:** Biofuels ⊃ ethanol + biodiesel + biogas — terms are not interchangeable.

27.9 Biogas

Identity: Anaerobic digestion | CH_4 + CO_2 | GOBAR-Dhan / SATAT

- Biogas forms when organic matter digests **without oxygen**.
- Major constituents: **methane (~50–70%)** + **carbon dioxide (~25–45%)**.
- **Butane** and **carbon monoxide** are not major biogas components.
- Slurry is useful bio-fertiliser; SATAT promotes compressed biogas (CBG) (संपीडित बायोगैस).

PYQ — UPPCS Prelims 2025, Q28

Major constituents of biogas: (1) Methane (2) Butane (3) Carbon dioxide (4) Carbon monoxide

Options: A. 1 and 3 B. 3 and 4 C. 2 and 3 D. 1 and 2

Show answer

Ans: A — Methane + CO_2.

27.10 Bioethanol

Identity: C_2H_5OH from fermentation | Petrol blend | E10 achieved | E20 by 2025–26

- **Bioethanol** is alcohol produced mainly by fermenting sugarcane molasses (also maize, 2G waste).
- Blended with petrol: **E10** (10%) largely achieved; **E20 by 2025–26** under Biofuels Policy 2018.
- Reduces petrol import bill and transport CO_2 intensity.
- Western UP sugar belt distilleries are key producers.

> **Logic:** Bioethanol = **petrol** engines; biodiesel = **diesel** engines.

27.11 Biodiesel

Identity: Transesterification | Jatropha/Pongamia/UCO | Diesel blend B5/B10

- **Biodiesel** is fatty acid methyl ester (**FAME**) made by **transesterification** of vegetable oils + methanol + catalyst → biodiesel + glycerol.
- Feedstocks: **Jatropha**, **Pongamia pinnata (karanj)**, used cooking oil (**RUCO** initiative).
- **Pongamia** grows in arid / saline (लवणीय) / alkaline soils; seeds are lipid-rich with nearly half **oleic acid** — classic dual-statement biodiesel stem with Jatropha.
- Blended with diesel as **B5/B10**.
- Biodiesel ≠ biogas (liquid ester vs methane gas).
- First Indian airline biofuel ATF flight (Aug **2018**): **SpiceJet** (Dehradun (देहरादून)–Delhi).

> **Logic:** Jatropha was promoted as non-edible biodiesel crop — yield challenges limited scale.

27.12 Small Hydropower

Identity: ≤25 MW MNRE | Himalayan potential | Run-of-river preferred

- MNRE defines **small hydro as ≤25 MW**; above that = large hydro (conventional).
- Himalayan rivers have **enormous** hydro potential — Indian projects in UK, HP, Sikkim, Arunachal.
- Run-of-river SHP has lower submergence than large storage dams.
- **Ramganga (रामगंगा) (~198 MW)** is **large hydro**, not SHP.

> **Logic:** Himalayan potential true; "Nepal border means zero Indian hydro" false.

27.13 Tidal Energy

Identity: Moon's gravity | Khambhat highest | Predictable | No major Indian plant yet

- Tidal energy uses rise/fall of sea level driven mainly by the **Moon**.
- India potential: **Gulf of Khambhat** (highest range), **Gulf of Kutch**.
- Highly predictable vs wind/solar; capital cost and estuary (ज्वारनदमुख) impact are high.
- Tidal ≠ wave energy (waves = wind-driven surface motion).

27.14 Ocean Energy

Identity: Wave + OTEC + salinity + tidal stream | Broader than tidal alone

- **Ocean energy** includes **wave**, **OTEC**, **salinity gradient**, and **tidal stream** technologies.
- **OTEC:** uses ≥~20°C temperature difference between warm surface and cold deep water — tropical seas (Lakshadweep (लक्षद्वीप)/Andaman (अंडमान) potential).
- Wave energy: wind-driven surface waves — R&D stage on Indian coasts.
- Trap: "Ocean energy means only tidal" = **FALSE**.

Type	Driver
Tidal	Moon gravity
Wave	Wind on sea surface
OTEC	Warm–cold water ΔT

> **Logic:** Ocean energy ⊃ tidal, but is not limited to it.

27.15 Energy Security

Identity: Reliable affordable supply | Oil import risk | SPR | Renewables + biofuels buffer

- **Energy security** means reliable, affordable energy supply without crippling import or shortage risk.
- India imports ~**85%** crude oil — price and geopolitics shock risk.
- Coal is largely **domestic** — electricity security anchor.
- **Strategic Petroleum Reserve (SPR):** Visakhapatnam, Mangalore, Padur — crisis oil buffer.
- Tools: diversify suppliers, expand renewables, biofuel blending, gas pipelines, grid reliability.
- UP has little fossil production — depends on inter-state power/fuel; local solar + ethanol help.

> **Logic:** Energy security ≠ oil alone — includes coal, gas, electricity, and strategic stocks.

27.16 Renewable Energy Scenario in India

Identity: MNRE nodal | Solar largest growth | Wind second | 500 GW non-fossil 2030

- MNRE is nodal for new and renewable energy (large hydro usually under Power Ministry).
- Rough installed picture (mid-2020s order): solar leading growth; wind second; biomass/SHP smaller shares.
- Earlier 175 GW by 2022 ambition expanded into **500 GW non-fossil by 2030**.
- Challenges: land, grid evacuation, storage, seasonal mismatch, state DISCOM finances.

Source	Key fact
Solar	Fastest capacity growth
Wind	TN/Gujarat strongholds
SHP	Himalayan streams ≤25 MW
Biomass	Bagasse, biogas, waste-to-energy

27.17 National Renewable Energy Programme

Identity: MNRE umbrella | NSM + wind + biomass + SHP + waste | PLI manufacturing

- National renewable push is an **MNRE programme cluster**, not one single Act.
- Includes National Solar Mission, wind policies, biomass/waste-to-energy, SHP support, offshore wind policy.
- **PLI for solar PV** supports domestic manufacturing (Atmanirbhar).
- Convergence with Power Ministry grid reforms and state renewable purchase obligations (RPOs).

27.18 National Solar Mission

Identity: 2010 NAPCC mission | Solar scale-up | Part of 8 National Missions

- Launched **2010** under **NAPCC (2008)** as one of eight National Missions.
- Aimed to make India a global solar leader — capacity, cost reduction, domestic manufacturing.
- Target path: early 20 GW ambition → revised to **100 GW solar** → contributes to 500 GW non-fossil 2030.
- Complements ISA (2015) and PM-KUSUM (2019).

> **Logic:** NSM = **2010**, under NAPCC — not 2015 (ISA year).

27.19 Energy Conservation

Identity: EC Act 2001 | BEE | Star labeling | Avoided demand

- **Energy conservation** means using less energy for the same service — avoid waste.
- **Energy Conservation Act 2001** created **Bureau of Energy Efficiency (BEE)** (ऊर्जा दक्षता ब्यूरो) under Ministry of Power.
- BEE sets standards, labels, and norms for designated consumers.
- **Star labeling (1–5 stars)** on ACs, fridges, fans, TVs, geysers — higher stars = lower consumption.
- Building energy codes and awareness campaigns cut demand without new generation.

Item	Year/body
Energy Conservation Act	**2001**
BEE	Under Power Ministry
Star labeling	Appliance efficiency

> **Logic:** BEE = efficiency; MNRE = renewables — different mandates.

27.20 Energy Efficiency

Identity: Same output, less input | PAT ESCerts | UJALA LEDs | PAT ≠ CCTS

- **Energy efficiency** means delivering the same service with less energy input.
- **PAT (Perform, Achieve and Trade (पण्याध्यक्ष)), 2012:** market scheme for energy-intensive industries — surplus efficiency becomes tradable **ESCerts**.
- PAT is **not** carbon credit (कार्बन क्रेडिट) trading (**CCTS 2023** is the carbon market).
- **UJALA:** mass LED distribution — large residential efficiency gain.
- Efficiency + renewables together cut emissions faster than either alone.

Scheme	What it trades/does
PAT	ESCerts (energy efficiency)
CCTS	Carbon credits (CO_2e)
UJALA	LED bulbs at scale

> **Logic:** PAT = **energy efficiency certificates**, not carbon credits.